

Enriched Finitely Presentable Banach Spaces Are Exactly the Finite-Dimensional Ones

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Status. Candidate full solution, likely valid, requiring expert review. A single fixed sequential diagram on c_0 detects every infinite-dimensional Banach space. Josefson–Nissenzweig supplies the obstructing morphism, and the left-shift tails violate the exact hom-metric condition for enriched finite presentability.

1 Source and exact question

Let **Ban** be the category of real or complex Banach spaces and linear contractions, enriched over complete metric spaces by the operator-norm metric on each hom-set. In Jiří Rosický, *Enriched purity and presentability in Banach spaces*, arXiv:2206.08546 and *Communications in Algebra* 51 (2023), 5242–5262 [1], Theorem 3.1 proves that every finite-dimensional Banach space is finitely presentable in this enriched sense.

Remark 3.3 then asks for the converse: must every enriched finitely presentable Banach space be finite-dimensional? The answer is yes. In fact, the converse is witnessed by one countable diagram, independent of the Banach space.

We say that a **CMet**-enriched category is *locally finitely presentable* if it has all weighted colimits and a set $\mathcal{A}$ of finitely presentable objects (in the enriched sense) such that every object is a directed colimit of objects of $\mathcal{A}$.

These categories are locally finitely presentable in the sense of [18] because they have a strong generator consisting of finitely presentable objects. However, **CMet** is locally finitely presentable in the sense of [18] (see (3.4) in this paper) but not locally finitely presentable in our sense.

Corollary 3.2. ***Ban** is locally finitely presentable as a category enriched over **CMet**.*

Proof. **Ban** has all weighted colimits (see [4] 4.5(3)). Thus the result follows from [3.1] and the fact that every Banach space is a directed colimit of finite-dimensional spaces. $\square$

Remark 3.3. [4] 7.7] characterizes Banach spaces A such that $\mathbf{Ban}(A, -) : \mathbf{Ban} \rightarrow \mathbf{CMet}$ preserves directed colimits of isometries as those admitting for every $\varepsilon > 0$ a morphism $u : A \rightarrow A_0$ to a finite-dimensional Banach space with $r : A_0 \rightarrow A$ such that $ru \sim_\varepsilon \text{id}_A$. These Banach spaces do not need to be finite-dimensional and satisfy (1) above for every directed colimit $(k_i : K_i \rightarrow K)$. Indeed, for $f : A \rightarrow K$ and $\varepsilon > 0$, we take $u : A \rightarrow A_0$ and $r : A_0 \rightarrow A$ with A_0 finite-dimensional and $ru \sim_{\frac{\varepsilon}{2}} \text{id}_A$. Following [3.1] there is $g : A_0 \rightarrow K_i$ such that $k_i g \sim_{\frac{\varepsilon}{2}} fr$. Then

$$k_i g u \sim_{\frac{\varepsilon}{2}} f r u \sim_{\frac{\varepsilon}{2}} f.$$

Hence $k_i g u \sim_\varepsilon f$.

We do not know whether these Banach spaces A can satisfy (2) as well. So, we do not know whether finitely presentable Banach spaces are finite-dimensional.

Proposition 3.4. *Let $U : \mathcal{K} \rightarrow \mathcal{L}$ and $F : \mathcal{L} \rightarrow \mathcal{K}$ be an enriched adjunction (where F is a left adjoint) such that U preserves directed colimits. Then F preserves finitely presentable objects (in the enriched sense).*

Proof. Let A be finitely presentable in $\mathcal{L}$. Since

$$\mathcal{K}(FA, -) \cong \mathcal{L}(A, U-)$$

and $\mathcal{L}(A, U-)$ preserves directed colimits, FA is finitely presentable. $\square$

4. PURITY

A morphism $f : K \rightarrow L$ in an ordinary locally finitely presentable category $\mathcal{K}$ is pure if in every commutative square

$$\begin{array}{ccc} K & \xrightarrow{f} & L \\ u \uparrow & & \uparrow v \\ A & \xrightarrow{g} & B \end{array}$$

Figure 1: Source page 8. Remark 3.3 states the finite-dimensionality question after recalling the two hom-metric conditions for preservation of directed colimits.

2 Full result

Write $c_0 = c_0(\mathbb{F})$ and let

$$S : c_0 \longrightarrow c_0, \quad S(a_1, a_2, a_3, \dots) = (a_2, a_3, a_4, \dots)$$

be the left shift. Consider the fixed sequential directed diagram

$$\mathcal{D} : \quad c_0 \xrightarrow{S} c_0 \xrightarrow{S} c_0 \xrightarrow{S} \dots \quad (1)$$

Theorem 1 (Classification and a universal detector). *For a real or complex Banach space A , the following are equivalent.*

- (i) A is finite-dimensional.
- (ii) A is finitely presentable in the **CMet**-enriched category **Ban**.
- (iii) The enriched representable functor

$$\mathbf{Ban}(A, -) : \mathbf{Ban} \longrightarrow \mathbf{CMet}$$

preserves the colimit of the single diagram $\mathcal{D}$ in (1).

Thus Remark 3.3 has a positive answer. The strengthening in (iii) says that no diagram depending on A is needed.

3 The two ingredients

Lemma 2 (The shift diagram has zero colimit). *The colimit of $\mathcal{D}$ in **Ban** is the zero Banach space.*

Proof. The zero maps form a cocone. Let $(u_n : c_0 \rightarrow E)_{n \geq 0}$ be any other cocone of contractions, so

$$u_n = u_{n+m} S^m \quad (n, m \geq 0).$$

For $z \in c_0$,

$$\|u_n z\| = \|u_{n+m} S^m z\| \leq \|S^m z\| \longrightarrow 0 \quad (m \rightarrow \infty).$$

Hence every u_n is zero. The zero cocone is therefore universal. $\square$

Lemma 3 (An infinite-dimensional witness). *If A is infinite-dimensional, there is a contraction $J : A \rightarrow c_0$ such that*

$$\|S^m J\| = 1 \quad (m \geq 0). \quad (2)$$

Proof. The Josefson–Nissenzweig theorem gives a sequence $(\varphi_r)_{r \geq 1} \subset A^*$ such that

$$\|\varphi_r\| = 1 \quad \text{for every } r, \quad \varphi_r(x) \longrightarrow 0 \quad \text{for every } x \in A$$

[2, 3]. Define

$$Jx = (\varphi_1(x), \varphi_2(x), \dots).$$

Pointwise nullity makes $Jx \in c_0$, and

$$\|J\| = \sup_{r \geq 1} \|\varphi_r\| = 1.$$

For every $m \geq 0$,

$$\|S^m J\| = \sup_{\|x\| \leq 1} \sup_{r > m} |\varphi_r(x)| = \sup_{r > m} \|\varphi_r\| = 1,$$

which is (2). $\square$

THE JOSEFSON–NISSENZWEIG PROPERTY FOR LOCALLY CONVEX SPACES

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ABSTRACT. We define a locally convex space E to have the *Josefson–Nissenzweig property* (JNP) if the identity map $(E', \sigma(E', E)) \rightarrow (E', \beta^*(E', E))$ is not sequentially continuous. By the classical Josefson–Nissenzweig theorem, every infinite-dimensional Banach space has the JNP. A characterization of locally convex spaces with the JNP is given. We thoroughly study the JNP in various function spaces. Among other results we show that for a Tychonoff space X , the function space $C_p(X)$ has the JNP iff there is a weak* null-sequence $(\mu_n)_{n \in \omega}$ of finitely supported sign-measures on X with unit norm. However, for every Tychonoff space X , neither the space $B_1(X)$ of Baire-1 functions on X nor the free locally convex space $L(X)$ over X has the JNP.

1. INTRODUCTION

All locally convex spaces (lcs for short) are assumed to be Hausdorff and infinite-dimensional, and all topological spaces are assumed to be infinite and Tychonoff. We denote by E' the topological dual of an lcs E . The dual space E' of E endowed with the weak* topology $\sigma(E', E)$ and the strong topology $\beta(E', E)$ is denoted by E'_{w^*} and E'_β , respectively. For a bounded subset $B \subseteq E$ and a functional $\chi \in E'$, we put $\|\chi\|_B := \sup\{|\chi(x)| : x \in B \cup \{0\}\}$.

Josefson [17] and Nissenzweig [20] proved independently the following theorem (other proofs of this beautiful result were given by Hagler and Johnson [15] and Bourgain and Diestel [6]).

Theorem 1.1 (Josefson–Nissenzweig). *Let E be a Banach space. Then there is a null sequence $\{\chi_n\}_{n \in \omega}$ in E'_{w^*} such that $\|\chi_n\| = 1$ for every $n \in \omega$.*

Therefore the identity map $E'_{w^*} \rightarrow E'_\beta$ is not sequentially continuous for every Banach space. Recall that a function $f : X \rightarrow Y$ between topological spaces X and Y is called *sequentially continuous* if for any convergent sequence $\{x_n\}_{n \in \omega} \subseteq X$, the sequence $\{f(x_n)\}_{n \in \omega}$ converges in Y and $\lim_n f(x_n) = f(\lim_n x_n)$.

The Josefson–Nissenzweig theorem was extended to Fréchet spaces in [5].

Theorem 1.2 (Bonet–Lindström–Valdivia). *For a Fréchet space E , the identity map $E'_{w^*} \rightarrow E'_\beta$ is sequentially continuous if and only if E is a Montel space.*

Another extension of the Josefson–Nissenzweig theorem was provided by Bonet [4] and Lindström and Schlumprecht [18] who proved that a Fréchet space E is a Schwartz space if and only if every null sequence in E'_{w^*} converges uniformly to zero on some zero-neighborhood in E .

Studying the separable quotient problem for C_p -spaces and being motivated by the Josefson–Nissenzweig theorem, Banach, Kąkol and Śliwa introduced in [3] the Josefson–Nissenzweig property for the space $C_p(X)$ of continuous real-valued functions on a Tychonoff space X , endowed with the topology of pointwise convergence. Namely, they defined $C_p(X)$ to have the Josefson–Nissenzweig property (JNP) if the dual space $C_p(X)'$ of $C_p(X)$ contains a weak* null sequence $\{\mu_n\}_{n \in \omega}$ of finitely supported sign-measures on X such that $\|\mu_n\| := |\mu_n|(X) = 1$ for every $n \in \omega$. This definition implies that for any Tychonoff space X containing a non-trivial convergent sequence, the function space $C_p(X)$

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Figure 2: A modern statement of the Josefson–Nissenzweig theorem: Theorem 1.1 in Banach–Gabrielyan [3].

4 Proof of Theorem 1

The implication (i) $\Rightarrow$ (ii) is Rosický's Theorem 3.1 [1]. The implication (ii) $\Rightarrow$ (iii) is immediate from the definition of enriched finite presentability.

It remains to prove (iii) $\Rightarrow$ (i). We argue contrapositively. Suppose A is infinite-dimensional and choose J from Lemma 3. Applying $\mathbf{Ban}(A, -)$ to $\mathcal{D}$ gives the sequential diagram of complete metric spaces

$$\mathbf{Ban}(A, c_0) \xrightarrow{S \circ -} \mathbf{Ban}(A, c_0) \xrightarrow{S \circ -} \mathbf{Ban}(A, c_0) \xrightarrow{S \circ -} \cdots$$

For a directed colimit of metric spaces, the distance between two elements represented at the same stage is the infimum of their distances at all later stages (followed by the usual zero-distance quotient and completion). Consequently, the distance in this colimit between the stage-zero classes of J and 0 is

$$\inf_{m \geq 0} \|S^m J - S^m 0\| = \inf_{m \geq 0} \|S^m J\| = 1.$$

In particular, this metric colimit is not a singleton.

On the other hand, Lemma 2 says $\text{colim } \mathcal{D} = 0$ in $\mathbf{Ban}$, and hence

$$\mathbf{Ban}(A, \text{colim } \mathcal{D}) = \mathbf{Ban}(A, 0)$$

is a singleton. The canonical comparison from the preceding metric colimit to $\mathbf{Ban}(A, 0)$ collapses the two points represented by J and 0 , which remain distance one apart. It is not an isomorphism in $\mathbf{CMet}$. Thus $\mathbf{Ban}(A, -)$ does not preserve this colimit, contradicting (iii). Therefore A is finite-dimensional.

Remark 4 (The source's condition (2) fails as sharply as possible). *Let $k_0 : c_0 \rightarrow 0$ be the colimit map. Taking $f = J$ and $g = 0$ at stage zero, the equality required in condition (2) on source page 6 is*

$$\|k_0 f - k_0 g\| = \inf_{m \geq 0} \|S^m f - S^m g\|.$$

The left side is 0, while the right side is 1. There is no limiting or completion subtlety: the defect is maximal and persists at every stage.

5 Why the shift is the exact detector

The finite-dimensional and infinite-dimensional cases are separated by uniformity on the unit ball.

If A is finite-dimensional and $T : A \rightarrow c_0$ is bounded, then $S^m T x \rightarrow 0$ for each x . On the compact unit ball of A , the continuous functions

$$q_m(x) = \|S^m T x\|$$

decrease pointwise to zero. Dini's theorem gives $\|S^m T\| = \sup_{\|x\| \leq 1} q_m(x) \rightarrow 0$. Thus every pair of morphisms becomes identified in the $\mathbf{CMet}$ -colimit of the represented shift diagram.

For infinite-dimensional A , Josefson–Nissenzweig is precisely the failure of that uniformity: the pointwise-null sequence (φ_r) has unit norms. Packaging the sequence as $J : A \rightarrow c_0$ converts nonuniform weak-star nullity into the persistent tail equality $\|S^m J\| = 1$. This explains why the same diagram works for all A and why neither separability nor a basis nor an approximation property is required.

6 Proof and upgrade audit

Eight routes or possible failure modes were checked.

- Route 1.** The Josefson–Nissenzweig sequence was packaged as an operator into c_0 and paired with the left shift. This gives the full solution.
- Route 2.** A scalar contraction chain was tested first. Although its Banach colimit is zero, the norms of every represented morphism also decay, so it cannot violate the hom-metric equality.
- Route 3.** Tail projections associated to a Schauder basis give the right phenomenon for spaces with a basis, but do not cover arbitrary Banach spaces. Josefson–Nissenzweig removes this restriction.
- Route 4.** A basic-sequence construction was tested. It lives naturally on a subspace and requires noncanonical extension control; the global weak-star null sequence is the correct replacement.
- Route 5.** Quotients by increasing finite-dimensional pieces were considered. They do not give a universal sequential exhaustion in nonseparable spaces. The fixed c_0 shift avoids all density-character issues.
- Route 6.** The zero-colimit claim was checked directly by the universal property, not merely by the usual seminorm formula.
- Route 7.** The **CMet** colimit was checked at the level of its exact pseudometric: J and 0 retain distance one before quotienting and completion, so they cannot be accidentally identified.
- Route 8.** Both real and complex scalars were audited. Josefson–Nissenzweig, $c_0(\mathbb{F})$, the shift, and the norm computation work in either case.

7 Novelty bounds and recommendation

The four lightweight run indexes were searched for arXiv:2206.08546, the paper title, enriched finite presentability, the exact Remark 3.3 phrasing, Josefson–Nissenzweig, and the c_0 shift; no prior run result was found.

A bounded Crossref/OpenAlex audit identified the published source DOI <https://doi.org/10.1080/00927872.2023.2228412> and exactly two citing works as of 11 August 2026. Chirvasitu [4] gives an alternative proof of the finite-dimensional positive direction, but does not discuss the converse. Rosický–Tendas [5] develops notions of enriched purity, but likewise gives no finite-presentability classification. Full texts of both citing papers were searched for the exact question, finite-dimensional converses, Josefson–Nissenzweig, and shift arguments. Bounded web/arXiv queries found the source and the standard Josefson–Nissenzweig literature, but no direct answer. This is not an exhaustive literature certification.

Human review should focus on only three points: the universal property in Lemma 2, the exact tail norm in Lemma 3, and the standard directed-colimit metric formula used in the final comparison. If accepted, the argument is a short full answer to Remark 3.3 and the fixed-diagram strengthening should be recorded with it.

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